

Pedestrians Movement on Crossings

Proof of Concept

Minimal working prototype of the pedestrian simulation

Course: Agent Systems

Authors:

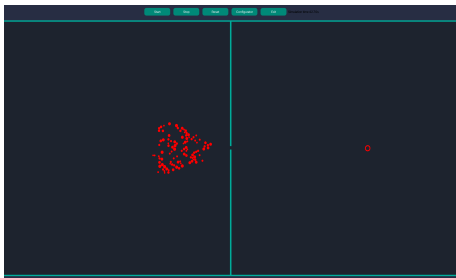
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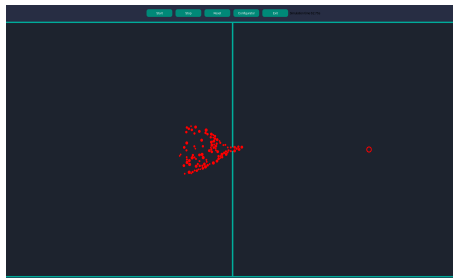
1. Result

The simulation engine and the Social Force based physical model have been successfully implemented and run end-to-end on a single bottleneck map. Agents are spawned on the left, navigate around a vertical wall and reach the destination on the right. The frames below show one full run captured at increasing simulation times.

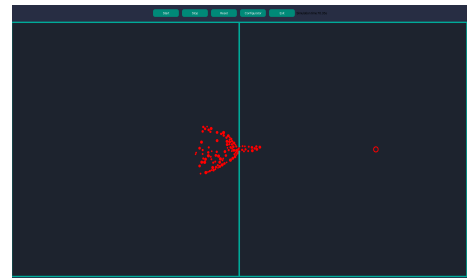
2. Screenshots



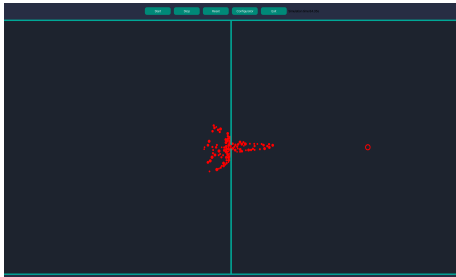
$t \approx 43 \text{ s}$



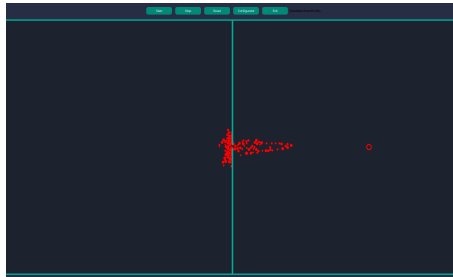
$t \approx 63 \text{ s}$



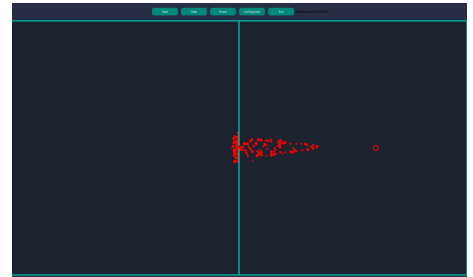
$t \approx 70 \text{ s}$



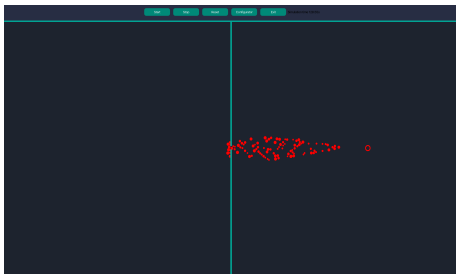
$t \approx 84 \text{ s}$



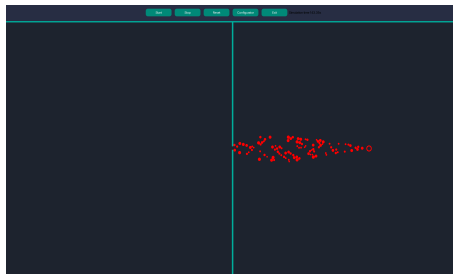
$t \approx 96 \text{ s}$



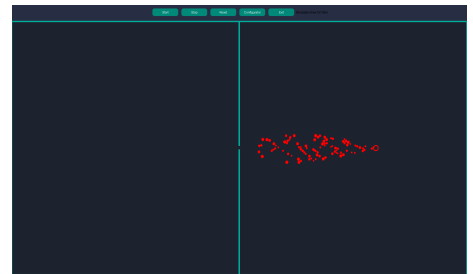
$t \approx 109 \text{ s}$



$t \approx 129 \text{ s}$



$t \approx 143 \text{ s}$



$t \approx 158 \text{ s}$

3. Next steps

- Application menu and runtime controls.
- Additional map layouts (crossings, corridors, multi-group scenarios).
- Parallel computation of agent updates (multi-threaded engine).
- Remaining features from the Agent Definition and State of the Art documents (crowd pressure visualisation, lane formation, freezing-by-heating validation).